

NLP Team

Executive Summary

StockSense is a financial intelligence platform that turns unstructured earnings call transcripts into clear, useful insights for investors and learners. Earnings calls contain important qualitative information such as management tone, strategy, risks, and future outlook, but this information is difficult to analyze because it exists only in long text form. Most current financial tools focus mainly on numbers like revenue and profit, while ignoring the valuable meaning hidden in how executives speak.

StockSense solves this problem by scraping earnings call transcripts and using AI to analyze management sentiment, extract key insights, track company performance over time, and provide Retrieval-Augmented Generation (RAG) based question answering. Users can ask questions such as “How is this company doing?” or “Is management confident?” and receive clear, data-backed explanations.

The main users of StockSense include retail investors, financial analysts, portfolio managers, fintech researchers, and business students. By reducing information overload and making qualitative financial data easier to understand, StockSense helps users make better investment and research decisions. The expected impact is improved transparency, stronger decision-making, and wider access to financial intelligence for non-expert users.

Problem Definition

Earnings calls are one of the most information-dense sources in financial markets. Beyond numerical performance, they contain management explanations, strategic intentions, risk disclosures, forward guidance, and narrative framing that directly shape investor expectations and market reactions. However, this information exists almost entirely in unstructured textual form, making it extremely difficult to analyze systematically, compare across time, or integrate into data-driven decision processes. Unlike financial statements, earnings calls are not standardized. Language varies across executives, companies, and industries. Tone, confidence, hesitation, ambiguity, and framing strategies are subjective and subtle, yet they often contain early signals of operational risk, strategic shifts, or future underperformance. Currently, these qualitative signals remain largely underutilized because they cannot be easily quantified, structured, or modeled at scale. As a result, market participants rely on manual reading, intuition, and selective interpretation, which introduces bias, inconsistency, and information loss. This problem affects multiple stakeholder groups:

- Retail investors lack the tools and expertise to interpret management language beyond surface-level summaries.
- Financial analysts spend significant time manually reading and summarizing transcripts, creating inefficiencies and scalability issues.
- Portfolio managers struggle to systematically integrate qualitative signals into quantitative investment strategies.
- Students and researchers lack structured datasets for large-scale narrative finance and NLP research.
- Fintech developers face high barriers in building products that leverage unstructured financial text.

This gap is important because research shows that language and tone in earnings calls are linked to future performance, risk, and investor confidence. Qualitative signals in earnings calls are strongly related to:

- Future earnings volatility
- Corporate risk exposure
- Strategic execution credibility
- Market confidence and investor trust
- Long-term firm valuation

When this information remains unstructured, it creates information gaps, increases bias, and weakens decision quality. As financial systems become more automated and data-driven, the inability to analyze corporate narratives represents a major weakness. StockSense addresses this problem by turning complex earnings call language into structured, usable financial intelligence.

Proposed Solution

StockSense is designed as an end-to-end AI-driven earnings call intelligence platform that converts unstructured corporate communication into structured, analyzable, and queryable financial intelligence. The system bridges the gap between qualitative corporate narratives and data-driven financial decision-making by transforming raw earnings call transcripts into machine-readable knowledge, analytical signals, and interactive insights.

Rather than acting as a simple transcript processing tool, StockSense functions as a financial narrative intelligence layer that integrates natural language processing, machine learning, and retrieval-augmented generation (RAG) into a unified platform. It enables users to move from passive reading of transcripts to active analytical reasoning over corporate narratives.

The system is organized into five tightly integrated functional modules:

1. **Transcript Ingestion & Structuring Layer**
This module automates the collection and standardization of earnings call data, ensures all downstream analytics operate on clean, structured, and standardized textual data.
2. **Management Tone & Sentiment Intelligence**
This module extracts psychological and linguistic signals from management communication. Instead of simple polarity classification, this module focuses on behavioral finance-oriented linguistic features that reflect managerial credibility, risk posture, and strategic confidence.
3. **Key Insight & Narrative Extraction Engine**
This module converts raw language into structured business intelligence. The output is a structured representation of corporate narratives that can be stored, queried, and compared.
4. **Longitudinal Performance & Narrative Analytics**
This module enables temporal and comparative analysis. This allows users to move beyond single-call analysis to longitudinal corporate behavior modeling.
5. **RAG-Based Intelligent Q&A System**
This module enables natural language interaction with corporate intelligence. Users can query the system in natural language while maintaining factual grounding through retrieval-augmented generation.

StockSense supports multiple high-value use cases:

- Investor analysis: “How has management confidence evolved over the past 6 quarters?”
- Risk assessment: “Which companies show increasing uncertainty before earnings declines?”
- Strategic research: “How did strategic priorities change after revenue contraction?”
- Comparative analysis: “Compare narrative risk signals between competitors.”

StockSense delivers value at multiple levels:

- Analytical value: Converts language into measurable signals
- Operational value: Reduces manual analysis workload
- Strategic value: Enhances qualitative decision intelligence
- Educational value: Enables narrative finance research
- Platform value: Creates reusable financial language infrastructure

Project success will be evaluated through:

- Accuracy of sentiment and tone classification
- Quality and relevance of extracted insights
- Consistency of longitudinal trend detection
- Reliability of RAG-based responses
- User query success rate

- Explainability and transparency of outputs
- User satisfaction and usability testing

Technical Approach

- Data Ingestion: Responsible for automated data acquisition and standardization.
 - Core Concepts: Data pipelines, ETL processes, data versioning, automation
 - Function: Automated scraping and API-based ingestion of earnings call transcripts, Scheduled data pipelines, Metadata enrichment (company, fiscal period, sector, event type), Version control for transcript updates
- Data Processing: Transforms raw text into structured machine-readable formats.
 - Core Concepts: Data cleaning, schema design, normalization, feature engineering
 - Function: Text cleaning and normalization, Speaker role classification, Dialogue segmentation, Noise filtering, Structured schema construction
- NLP Intelligence Layer: Performs semantic and linguistic modeling.
 - Core Concepts: NLP modeling, supervised learning, unsupervised learning, semantic analysis
 - Function: Finance-domain sentiment modeling, Confidence/uncertainty classification, Risk language detection, Topic modeling, Named Entity Recognition (NER), Automated summarization
- Embedding & Knowledge Layer: Transforms processed text into vectorized knowledge.
 - Core Concepts: Representation learning, semantic embeddings, vectorization, knowledge modeling
 - Function: Semantic embedding generation, Multi-resolution vectorization (sentence, paragraph, document), Temporal tagging, Knowledge indexing
- Retrieval Layer: Provides efficient access to knowledge.
 - Core Concepts: Information retrieval, similarity search, ranking algorithms, hybrid retrieval
 - Function: Vector database indexing, Hybrid retrieval (semantic + keyword), Context ranking and relevance scoring, Evidence selection
- RAG Reasoning Layer: Enables grounded generative intelligence. This layer ensures AI responses are evidence-based and verifiable.
 - Core Concepts: Retrieval-Augmented Generation, grounded generation, explainable AI
 - Function: Context injection into LLMs, Evidence-grounded response generation, Hallucination mitigation, Source attribution
- Application Layer: Provides user-facing functionality.
 - Core Concepts: Human–AI interaction, data visualization, decision support systems
 - Function: Interactive dashboards, Natural language query interface, Time-series visualization, Comparative analytics views, Explainability panels

The technical architecture of StockSense is not a collection of isolated tools, but a coherent AI system in which each layer maps academic concepts to production-grade technologies. This design enables:

- End-to-end transformation of unstructured financial language into structured intelligence
- Seamless integration between NLP modeling, retrieval systems, and generative AI
- Scalable data engineering pipelines
- Evidence-grounded AI reasoning through RAG
- Transparent and explainable decision support

By tightly coupling core analytical concepts with concrete system components, StockSense functions as a full-stack financial intelligence infrastructure rather than a standalone analytics tool. This architecture ensures extensibility, research value, and real-world applicability, making the platform suitable for both academic exploration and practical financial applications.

Challenges & Mitigation Strategies:

Challenge	Technical Risk	Mitigation Strategy
Transcript noise	Low model accuracy	Multi-stage preprocessing
Financial jargon	Semantic misinterpretation	Domain-tuned models
Model hallucination	Incorrect outputs	RAG grounding
Scalability	Performance bottlenecks	Modular architecture
Data drift	Model degradation	Continuous retraining

Team Plan

Roles

- Ziqi Huang(Peter): ML Engineer: NLP models, sentiment, embeddings
- Data Engineer: scraping, pipelines, storage
- Backend Developer: APIs, RAG system
- Xuanzhu Luo(Audrey): UI/UX, dashboards
- Project Lead: integration, planning, QA

Timeline

- Week 1–2: Data pipeline + scraping
- Week 3–4: NLP processing + sentiment models
- Week 5–6: Embeddings + vector DB
- Week 7–8: RAG integration
- Week 9: Frontend + UI
- Week 10: Testing + deployment

Communication Plan

- Weekly sprint meetings
- Shared documentation
- GitHub collaboration
- Task tracking (Notion/Jira)

Risk Management

- Model failure → fallback pipelines
- Data issues → redundant sources
- Scope creep → strict MVP definition
- Integration risk → modular architecture